

Electrical Power BOQ — POWER BOQ

Al Barsha South Third · Owner: APPROVAL · Consultant: APPROVAL WITHOUT ANY ADDITIONAL COST. 26.THE CONTRACTOR HAS TO PROVIDE NECESSARY ACCESSORIES TO COMPLETE THE INSTALLATIONS (WITHOUT ADDITIONAL FEES) · Date: 04/05/2026

1. PROJECT SUMMARY	
Field	Value
Project	—
Location	Al Barsha South Third
Owner / Client	APPROVAL
Consultant	APPROVAL WITHOUT ANY ADDITIONAL COST. 26.THE CONTRACTOR HAS TO PROVIDE NECESSARY ACCESSORIES TO COMPLETE THE INSTALLATIONS (WITHOUT ADDITIONAL FEES)
Date	04/05/2026
Authority	DEWA (Dubai Electricity & Water Authority)
Discipline	Electrical (Power) — single discipline
Building Total Connected Load	0.00 kW
Building Maximum Demand	~0 kW

2. INCOMING SUPPLY & TRANSFORMERS			
#	Description	Unit	Qty

4. SUB-MAIN DISTRIBUTION BOARDS (SMDB)				
#	Tag	Connected Load (kW)	Feeder Cable	Qty
4.1	SMDB-1F (1F)	150.86	1X4C 150mm2 XLPE/SWA/PVC +1x1x70mm2 ECC	1
4.2	SMDB-2F (2F)	150.86	1X4C 150mm2 XLPE/SWA/PVC +1x1x70mm2 ECC	1
4.3	SMDB-3F (3F)	150.86	1X4C 150mm2 XLPE/SWA/PVC +1x1x70mm2 ECC	1
4.4	SMDB-4F (4F)	150.86	1X4C 150mm2 XLPE/SWA/PVC +1x1x70mm2 ECC	1
4.5	SMDB-5F (5F)	150.86	1X4C 150mm2 XLPE/SWA/PVC +1x1x70mm2 ECC	1
4.6	SMDB-6F (6F)	150.86	1X4C 150mm2 XLPE/SWA/PVC +1x1x70mm2 ECC	1
4.7	SMDB-7F (7F)	150.86	1X4C 150mm2 XLPE/SWA/PVC +1x1x70mm2 ECC	1
4.8	SMDB-8F (8F)	150.86	1X4C 150mm2 XLPE/SWA/PVC +1x1x70mm2 ECC	1
4.9	SMDB-TF-odd (1F/3F/5F/7F)	150.86	1X4C 185mm2 XLPE/SWA/PVC +1C 95mm2 ECC	4
4.10	SMDB-TF-even (2F/4F/6F/8F)	150.86	1X4C 185mm2 XLPE/SWA/PVC +1C 95mm2 ECC	4
4.11	SMDB-RF (Roof)	248.63	2X4C 150mm2 XLPE/SWA/PVC +1X1C 150mm2 ECC	1
4.12	SMDB-G (Ground)	60.55	1X4C 50mm2 XLPE/SWA/PVC +1X1C 25mm2 ECC	1
4.13	SMDB-EV (Basement)	88.00	1X4C 70mm2 XLPE/SWA/PVC +1X1C 35mm2 ECC	1
4.14	SMDB-SH01 (Ground)	15.00	1X4C 10mm2 XLPE/SWA/PVC	1
4.15	SMDB-SH02 (Ground)	15.00	1X4C 10mm2 XLPE/SWA/PVC	1
4.16	SMDB-SH03 (Ground)	15.00	1X4C 10mm2 XLPE/SWA/PVC	1
4.17	SMDB-SH04 (Ground)	15.00	1X4C 10mm2 XLPE/SWA/PVC	1
4.18	SMDB-SH05 (Ground)	25.00	1X4C 10mm2 XLPE/SWA/PVC	1

4.19	SMDB-SH06 (Ground)	10.00	1X4C 10mm2 XLPE/SWA/PVC	1
4.20	SMDB-SH07 (Ground)	10.00	1X4C 10mm2 XLPE/SWA/PVC	1
4.21	SMDB-SH08 (Ground)	40.00	1X4C 10mm2 XLPE/SWA/PVC	1
4.22	SMDB-SH09 (Ground)	10.00	1X4C 10mm2 XLPE/SWA/PVC	1
4.23	SMDB-SH10 (Ground)	15.00	1X4C 10mm2 XLPE/SWA/PVC	1
4.24	SMDB-SH11 (Ground)	15.00	1X4C 10mm2 XLPE/SWA/PVC	1
4.25	SMDB-SH12 (Ground)	30.00	1X4C 16mm2 XLPE/SWA/PVC	1
4.26	ESMDB-G (Ground)	190.08	4C 70mm2 FIRE RATED +1C 35mm2 ECC	1
4.27	ESMDB-RF (Roof)	88.69	4C 70mm2 FIRE RATED +1C 35mm2 ECC	1

5. DISTRIBUTION BOARDS (DB)				
#	DB Tag	SMDB	Floor	Cable Size Qty
5.1	DB-G	SMDB-G	Ground	4C 16mm2 XLPE/SWA/ 1
5.2	DB-F1	SMDB-G	Ground	4C 10mm2 XLPE/SWA/ 1
5.3	DB-F4	SMDB-G	Ground	4C 10mm2 XLPE/SWA/ 1
5.4	DB-F5	SMDB-G	Ground	4C 10mm2 XLPE/SWA/ 1
5.5	DB-L.GYM	SMDB-RF	Roof	4C 10mm2 XLPE/SWA/ 1
5.6	DB-G.GYM	SMDB-RF	Roof	4C 10mm2 XLPE/SWA/ 1
5.7	DB-WM	SMDB-RF	Roof	4C 10mm2 XLPE/SWA/ 1
5.8	EDB-G	ESMDB-G	Ground	4C 10mm2 FIRE RATED 1
5.9	EDB-1F	ESMDB-G	1F	4C 10mm2 FIRE RATED 1
5.10	EDB-TEL	ESMDB-G	Ground	4C 10mm2 FIRE RATED 1
5.11	EDB-PR	ESMDB-G	Underground	4C 10mm2 FIRE RATED 1
5.12	EDB-4F	ESMDB-RF	4F	4C 10mm2 FIRE RATED 1
5.13	EDB-7F	ESMDB-RF	7F	4C 10mm2 FIRE RATED 1
5.14	EDB-GSM-R	ESMDB-RF	Roof	4C 10mm2 FIRE RATED 1
5.15	DB-T01 to DB-T15 (odd floors)	SMDB-TF-odd	1F/3F/5F/7F	4C 16mm2 XLPE/SWA/ 1
5.16	DB-T01 to DB-T15 (even floors)	SMDB-TF-even	2F/4F/6F/8F	4C 16mm2 XLPE/SWA/ 1
5.17	DB-SHOP01 to DB-SHOP12	SMDB-SH01 to SMDB-SH12	Ground	16mm2 SWA 1C 16mm2 ECC by tenant 1

6. MECHANICAL & SERVICE EQUIPMENT (Isolators + Dedicated Feeders)			
#	Description	Rating	Qty
6.1	Fire pump isolator (UAE Fire Code 2018)	15 kW	1
6.2	Jockey pump isolator	2.2 kW	1
6.3	Lift motor isolator	11 kW	2
6.4	Standby generator isolator (ATS-side)	—	1

7. POWER OUTLETS & ACCESSORIES			
#	Description	Unit	Est. Qty
7.1	13A single switched socket outlet	No.	~110

7.2	13A twin switched socket outlet	No.	~44
7.3	20A DP switched outlet (water heater)	No.	~22
7.4	FCU spur (per apartment)	No.	~33
7.5	USB charging outlet	No.	~11

8. CABLES (Main Distribution — measured from SLD; lengths via scaled floor plans)

#	Cable Specification	Application	Est. Length (m)
8.1	150mm ² XLPE	LVP-01 !' SMDB-1F (4C 150mm ² XLPE/ SWA/PVC +1x1x70mm ² ECC)	~25
8.2	150mm ² XLPE	LVP-01 !' SMDB-2F (4C 150mm ² XLPE/ SWA/PVC +1x1x70mm ² ECC)	~29
8.3	150mm ² XLPE	LVP-01 !' SMDB-3F (4C 150mm ² XLPE/ SWA/PVC +1x1x70mm ² ECC)	~33
8.4	150mm ² XLPE	LVP-01 !' SMDB-4F (4C 150mm ² XLPE/ SWA/PVC +1x1x70mm ² ECC)	~37
8.5	150mm ² XLPE	LVP-01 !' SMDB-5F (4C 150mm ² XLPE/ SWA/PVC +1x1x70mm ² ECC)	~41
8.6	150mm ² XLPE	LVP-01 !' SMDB-6F (4C 150mm ² XLPE/ SWA/PVC +1x1x70mm ² ECC)	~45
8.7	150mm ² XLPE	LVP-01 !' SMDB-7F (4C 150mm ² XLPE/ SWA/PVC +1x1x70mm ² ECC)	~49
8.8	150mm ² XLPE	LVP-01 !' SMDB-8F (4C 150mm ² XLPE/ SWA/PVC +1x1x70mm ² ECC)	~53
8.9	185mm ² XLPE	LVP-01 !' SMDB-TF-odd (1X4C 185mm ² XLPE/SWA/PVC +1C 95mm ² ECC)	~45
8.10	185mm ² XLPE	LVP-01 !' SMDB-TF-even (1X4C 185mm ² XLPE/SWA/PVC +1C 95mm ² ECC)	~45
8.11	150mm ² XLPE	LVP-02 !' SMDB-RF (2X4C 150mm ² XLPE/ SWA/PVC +1X1C 150mm ² ECC)	~65
8.12	50mm ² XLPE	LVP-02 !' SMDB-G (1X4C 50mm ² XLPE/ SWA/PVC +1X1C 25mm ² ECC)	~15
8.13	70mm ² XLPE	LVP-02 !' SMDB-EV (1X4C 70mm ² XLPE/ SWA/PVC +1X1C 35mm ² ECC)	~20
8.14	10mm ² XLPE	LVP-02 !' SMDB-SH01 (1X4C 10mm ² XLPE/SWA/PVC)	~20
8.15	10mm ² XLPE	LVP-02 !' SMDB-SH02 (1X4C 10mm ² XLPE/SWA/PVC)	~22
8.16	10mm ² XLPE	LVP-02 !' SMDB-SH03 (1X4C 10mm ² XLPE/SWA/PVC)	~24
8.17	10mm ² XLPE	LVP-02 !' SMDB-SH04 (1X4C 10mm ² XLPE/SWA/PVC)	~26
8.18	10mm ² XLPE	LVP-02 !' SMDB-SH05 (1X4C 10mm ² XLPE/SWA/PVC)	~28
8.19	10mm ² XLPE	LVP-02 !' SMDB-SH06 (1X4C 10mm ² XLPE/SWA/PVC)	~30
8.20	10mm ² XLPE	LVP-02 !' SMDB-SH07 (1X4C 10mm ² XLPE/SWA/PVC)	~32
8.21	10mm ² XLPE	LVP-02 !' SMDB-SH08 (1X4C 10mm ² XLPE/SWA/PVC)	~34
8.22	10mm ² XLPE	LVP-02 !' SMDB-SH09 (1X4C 10mm ² XLPE/SWA/PVC)	~36
8.23	10mm ² XLPE	LVP-02 !' SMDB-SH10 (1X4C 10mm ² XLPE/SWA/PVC)	~38
8.24	10mm ² XLPE	LVP-02 !' SMDB-SH11 (1X4C 10mm ² XLPE/SWA/PVC)	~40
8.25	16mm ² XLPE	LVP-02 !' SMDB-SH12 (1X4C 16mm ² XLPE/SWA/PVC)	~42
8.26	70mm ² FIRE_RATED	LVP-02 !' ESMDB-G (4C 70mm ² FIRE RATED +1C 35mm ² ECC)	~15
8.27	70mm ² FIRE_RATED	ESMDB-G !' ESMDB-RF (4C 70mm ² FIRE RATED +1C 35mm ² ECC via 225mm	~65
8.28	185mm ² FIRE_RATED	LVP-02 !' Fire Pump (1X4C 185mm ² FP400 FIRE RATED +1C 95mm ² FR	~35
8.29	16mm ² XLPE	SMDB-SHA DB (4C 16mm ² XLPE/SWA/ PVC +1C 16mm ² ECC)	~20
8.30	10mm ² XLPE	SMDB-G !' DB-F1 (4C 10mm ² XLPE/SWA/ PVC +1C 10mm ² ECC)	~15
8.31	10mm ² XLPE	SMDB-G !' DB-F4 (4C 10mm ² XLPE/SWA/ PVC +1C 10mm ² ECC)	~15
8.32	10mm ² XLPE	SMDB-G !' DB-F5 (4C 10mm ² XLPE/SWA/ PVC +1C 10mm ² ECC)	~15
8.33	10mm ² XLPE	SMDB-RF !' DB-L.GYM (4C 10mm ² XLPE/ SWA/PVC +1C 10mm ² ECC)	~20

8.34	10mm² XLPE	SMDB-RF !' DB-G.GYM (4C 10mm2 XLPE/ SWA/PVC +1C 10mm2 ECC)	~20
8.35	10mm² XLPE	SMDB-RF !' DB-WM (2C 10mm2 XLPE/ SWA/PVC +1C 10mm2 ECC)	~15
8.36	10mm² FIRE_RATED	ESMDB-G !' EDB-G (4C 10mm2 FIRE RATED +1C 10mm2 ECC)	~15
8.37	6mm² FIRE_RATED	ESMDB-G !' EDB-1F (4C 6mm2 FIRE RATED +1C 6mm2 ECC)	~25
8.38	10mm² FIRE_RATED	ESMDB-G !' EDB-TEL (4C 10mm2 FIRE RATED +1C 10mm2 ECC)	~15
8.39	10mm² FIRE_RATED	ESMDB-G !' EDB-PR (4C 10mm2 FIRE RATED +1C 10mm2 ECC)	~30
8.40	10mm² FIRE_RATED	ESMDB-RF !' EDB-4F (4C 10mm2 FIRE RATED +1C 10mm2 ECC)	~20
8.41	10mm² FIRE_RATED	ESMDB-RF !' EDB-7F (4C 10mm2 FIRE RATED +1C 10mm2 ECC)	~10
8.42	10mm² FIRE_RATED	ESMDB-RF !' EDB-GSM-R (4C 10mm2 FIRE RATED +1C 10mm2 ECC)	~15
8.43	10mm² FIRE_RATED	ESMDB-RF !' Lift-01 (4C 10mm2 FIRE RATED +1C 10mm2 ECC, 40A TP)	~35
8.44	10mm² FIRE_RATED	ESMDB-RF !' Lift-02 (4C 10mm2 FIRE RATED +1C 10mm2 ECC, 40A TP)	~35
8.45	10mm² FIRE_RATED	ESMDB-RF !' Lift-03 (4C 10mm2 FIRE RATED +1C 10mm2 ECC, 40A TP)	~35
8.46	10mm² FIRE_RATED	ESMDB-RF !' Booster Pump (4C 10mm2 FIRE RATED +1C 10mm2 ECC, 40A TP)	~20
8.47	16mm² XLPE	SMDB-TF-odd !' DB-T01 to DB-T15 odd floors (4C 16mm2 XLPE/SWA/PVC +1C 16mm2 ECC)	~25
8.48	16mm² XLPE	SMDB-TF-even !' DB-T01 to DB-T15 even floors (4C 16mm2 XLPE/SWA/PVC +1C 16mm2 ECC)	~25

9. CONTAINMENT (Cable Tray, Trunking, Conduit)			
#	Description	Unit	Est. Qty
9.1	Cable tray (HDGI, 300mm)	m	~1527
9.2	Cable tray (HDGI, 200mm)	m	~1018
9.3	PVC conduit (25mm)	m	~1527

10. EARTHING & LIGHTNING PROTECTION			
#	Description	Unit	Qty
10.1	Earth pit assembly (DEWA approved, 2.4m rod)	No.	4
10.2	Earth cable 70mm² Cu/PVC	m	72
10.3	Surge protection device (Type 1+2)	No.	1

11. METERING & MONITORING			
#	Description		Qty
11.1	DEWA kWh meter (3-phase)		2
11.2	CT meter set		0

12. SUMMARY OF ELECTRICAL LOADS				
Panel	TCL (kW)	Standby (kW)	Demand Factor	Max Demand (kW)
—	0.00	0.00	0.00	0.00
—	0.00	0.00	0.00	0.00
—	0.00	0.00	0.00	0.00
—	0.00	0.00	0.00	0.00
—	0.00	0.00	0.00	0.00

—	0.00	0.00	0.00	0.00
—	0.00	0.00	0.00	0.00
TOTAL BUILDING	0.00	—	—	~0.00